

CS 305 Lab Tutorial

Lab13 VLAN, Switch

Dept. Computer Science and Engineering
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Topic

- **LAN(Local Area Network)**
 - **A LAN is typically a broadcast domain**
- **VLAN(Virtual Local Area Network)**
 - **Isolate broadcast domain**
- **Switch**
 - Physical interface
 - link-type: access vs trunk
 - VLAN interface
 - IP address
 - DHCP server
- Practices(page10-11, 14-15, 20)

Tips on eNSP : useful commands while in CLI

➤ **display**

- **dis this**
- **dis ip routing-table**
- **dis mac-address**
- **dis vlan <vlan_id>**

➤ **enable DHCP**

- **[Huawei] dhcp enable**

➤ **configure interface**

- **ip address <ip address> <subnet mask>**
- **port link-type access**
- **port default vlan <vlan_id>**
- **port link-type trunk**
- **port trunk allow-pass vlan 13 vlan24**

➤ **configure vlan**

- **vlan <vlan_id>**

➤ **configure vlan interface**

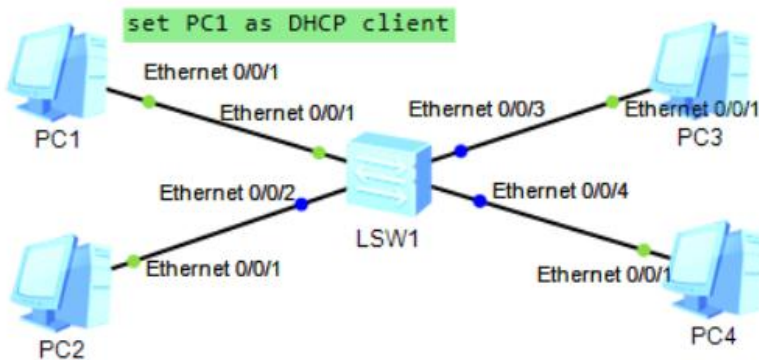
- **inter vlan <vlan_id>**
- **ip address <ip address> <subnet mask>**
- **dhcp select interface (optional)**

LAN-broadcast domain(1)

- **LAN(Local Area Network):**
 - A Local Area Network (LAN) is a network that covers a limited geographical area (such as an office building or campus), relies on high-speed physical channels for communication, and maintains a consistently low bit error rate.
 - **A LAN is typically a broadcast domain**, meaning any broadcast frame sent by a device within this network can be received by all other devices in the same domain.

Demo1. Build a simple LAN, with 4 PCs and a Switch.

- set PC1 as DHCP and DHCPv6 client
- Capture packets on the interface Ethernet 0/0/2, Ethernet 0/0/3 and Ethernet 0/0/4 of the switch.



PC1 Configuration Window:

Host Name:

MAC Address: 54-89-98-F3-70-DE

IPv4 Configuration:

- ☐ Static
- ☒ DHCP
- ☒ Obtain DNS server address automatically

IP Address:

Subnet Mask:

Gateway:

DNS1:

DNS2:

IPv6 Configuration:

- ☐ Static
- ☒ DHCPv6

IPv6 Address:

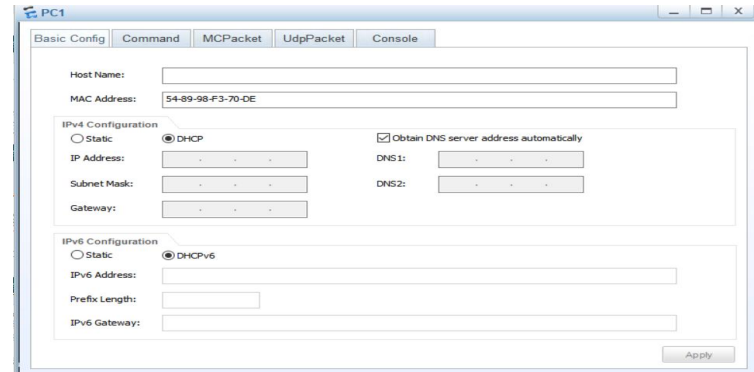
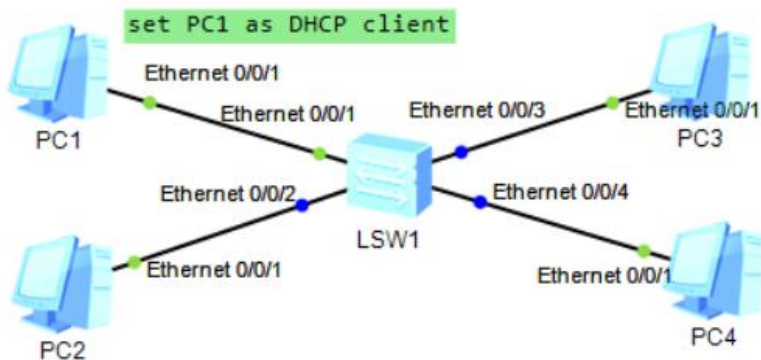
Prefix Length:

IPv6 Gateway:

Apply

LAN-broadcast domain(2)

Demo1. In the LAN, PC1 is set as a DHCP client, which means PC1 would invoke DHCP discover packets. The destination of DHCP discover packets is broadcast address.



The broadcast packets could be forward to all interfaces on the switch except for the receiving interface on.

The broadcast packets were captured on both Ethernet 0/0/2, Ethernet 0/0/3 and Ethernet 0/0/4 of the switch.

dhcp					
No.	Tim	Source	Protocol	Destination	Info
1...		0.0.0.0	DHCP	255.255.255.255	DHCP Discover - Transaction ID 0x...
<					
> Frame 78: 410 bytes on wire (3280 bits), 410 bytes captured (3280 bits) on interface -, id 0					
> Ethernet II, Src: HuaweiTe_f3:70:de (54:89:98:f3:70:de), Dst: Broadcast (ff:ff:ff:ff:ff:ff)					
> Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255					
> User Datagram Protocol, Src Port: 68, Dst Port: 67					
> Dynamic Host Configuration Protocol (Discover)					

dhcpv6					
No.	Tim	Source	Protocol	Destination	Info
1...		fe80::5689:98ff:f...	DHCPv6	ff02::1:2	Solicit XID: 0x38cdf0 CID: 00030001548998f370de
<					
> Frame 79: 112 bytes on wire (896 bits), 112 bytes captured (896 bits) on interface -, id 0					
> Ethernet II, Src: HuaweiTe_f3:70:de (54:89:98:f3:70:de), Dst: IPv6mcast 01:00:02 (33:33:00:01:00:02)					
> Internet Protocol Version 6, Src: fe80::5689:98ff:fef3:70de, Dst: ff02::1:2					
> User Datagram Protocol, Src Port: 546, Dst Port: 547					
> DHCPv6					

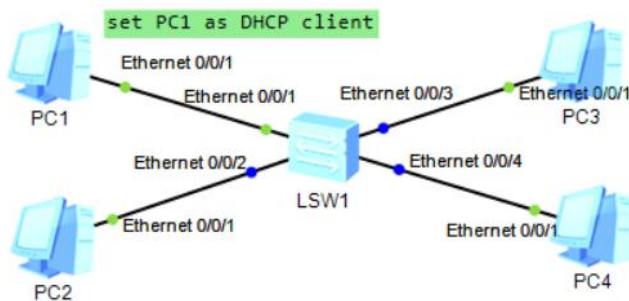
VLAN(1)-default vlan

VLAN(Virtual Local Area Network) is a communication technology that divides a physical LAN into multiple broadcast domains logically.

- The hosts in VLAN can communicate with each other directly
- The VLANs cannot communicate with each other directly, so the broadcast message is limited in one VLAN .

Demo1. Build a simple network topology, with 4 PCs and a Switch, PC1 is set as DHCP client.
Using “**display vlan 1**” to display the default vlan in the Switch(LSW1).

All the interface in the switch are in the vlan 1 by default.



LSW1

[Huawei]dis vlan 1

U: Up; D: Down; TG: Tagged; UT: Untagged;

MP: Vlan-mapping; ST: Vlan-stacking;

#: ProtocolTransparent-vlan; *: Management-vlan;

VID	Type	Ports
1	common	UT:GE0/0/1 (U)
		GE0/0/2 (U)
		GE0/0/3 (U)
		GE0/0/4 (U)
		GE0/0/5 (D)
		GE0/0/6 (D)
		GE0/0/7 (D)
		GE0/0/8 (D)
GE0/0/9 (D)		
GE0/0/10 (D)		
GE0/0/11 (D)		
GE0/0/12 (D)		
GE0/0/13 (D)		
GE0/0/14 (D)		
GE0/0/15 (D)		
GE0/0/16 (D)		
GE0/0/17 (D)		
GE0/0/18 (D)		
GE0/0/19 (D)		
GE0/0/20 (D)		
GE0/0/21 (D)		
GE0/0/22 (D)		
GE0/0/23 (D)		
GE0/0/24 (D)		

VID	Status	Property	MAC-LRN	Statistics	Description
1	enable	default	enable	disable	VLAN 0001

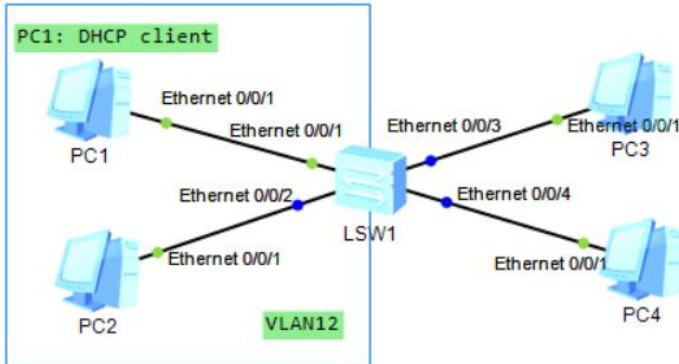
[Huawei]

VLAN(2)-create vlan

Demo2: Based on Demo1, make Eth 0/0/1 and Eth 0/0/2 of Switch(LSW1) access to vlan 12;

Configuration:

- step 1. create vlan (in the system view)
 - command: # to create one vlan
 - vlan <vlan-ID>
 - .e.g. **vlan 12**
- step 2. specify the interface which access to the vlan and the vlan id(in the interface configuration view)
 - command: **port link-type access**
 - command: **port default vlan 12**

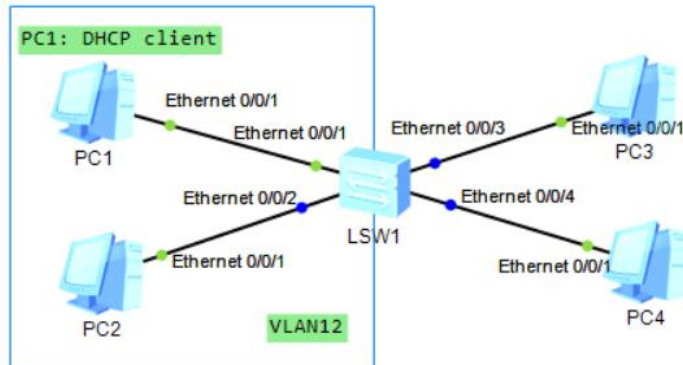


```
[Huawei]int eth 0/0/1
[Huawei-Ethernet0/0/1]dis this
#
interface Ethernet0/0/1
port link-type access
port default vlan 12
#
return
```

```
[Huawei]int eth 0/0/2
[Huawei-Ethernet0/0/2]dis this
#
interface Ethernet0/0/2
port link-type access
port default vlan 12
#
return
```


VLAN(3)-2 vlans(1)

Observing the state of VLAN using “**dis VLAN <vlan id>**”, it was found that: Interfaces Eth0/0/1 and Eth 0/0/2 are no longer part of the default VLAN 1 as they have entered VLAN 12.



```
LSW1
[Huawei]dis vlan 1
-----
U: Up;           D: Down;           TG: Tagged;       UT: Untagged;
MP: Vlan-mapping; ST: Vlan-stacking;
#: ProtocolTransparent-vlan; *: Management-vlan;
-----

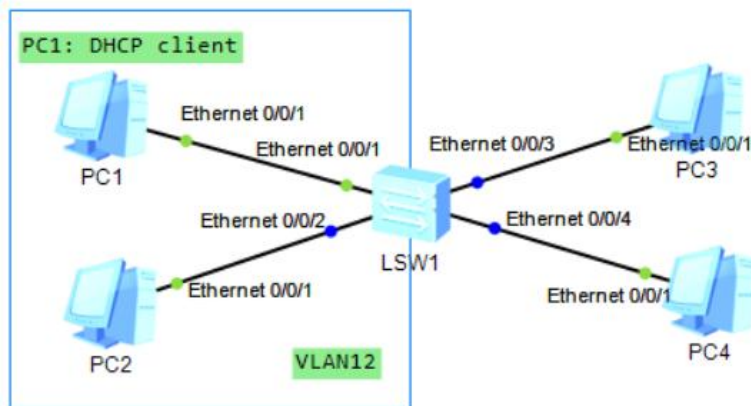
VID  Type  Ports
-----
1    common  UT:Eth0/0/3 (U)   Eth0/0/4 (U)   Eth0/0/5 (D)   Eth0/0/6 (D)
                        Eth0/0/7 (D)   Eth0/0/8 (D)   Eth0/0/9 (D)   Eth0/0/10 (D)
                        Eth0/0/11 (D)  Eth0/0/12 (D)  Eth0/0/13 (D)  Eth0/0/14 (D)
                        Eth0/0/15 (D)  Eth0/0/16 (D)  Eth0/0/17 (D)  Eth0/0/18 (D)
                        Eth0/0/19 (D)  Eth0/0/20 (D)  Eth0/0/21 (D)  Eth0/0/22 (D)
                        GE0/0/1 (D)   GE0/0/2 (D)

VID  Status  Property  MAC-LRN  Statistics  Description
-----
1    enable  default  enable  disable  VLAN 0001
[Huawei]dis vlan 12
-----

VID  Type  Ports
-----
12   common  UT:Eth0/0/1 (U)   Eth0/0/2 (U)

VID  Status  Property  MAC-LRN  Statistics  Description
-----
12   enable  default  enable  disable  VLAN 0012
[Huawei]
```


VLAN(4)- 2 vlans(2)



The DHCP request broadcast pkts sent by PC1 would only be received on interface Eth0/0/2, and CANNOT be received on the other two interfaces(Eth0/0/3 and Eth0/0/4) on the switch(LSW1).

The broadcast domain is isolated.

Wireshark capture showing a DHCP Discover packet (Transaction ID 0x00000000) sent by PC1 (Source: 0.0.0.0) to the switch (Destination: 255.255.255.255). The packet is captured on interface 0.

No.	Time	Source	Protocol	Destination	Info
1...	0.0.0.0	0.0.0.0	DHCP	255.255.255.255	DHCP Discover - Transaction ID 0x00000000

Frame 2: 410 bytes on wire (3280 bits), 410 bytes captured (3280 bits) on interface -, id 0
> Ethernet II, Src: HuaweiTe_f3:70:de (54:89:98:f3:70:de), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.255.255
> User Datagram Protocol, Src Port: 68, Dst Port: 67
> Dynamic Host Configuration Protocol (Discover)

Wireshark capture showing a DHCP Discover packet (Transaction ID 0x00000000) sent by PC1 (Source: 0.0.0.0) to the switch (Destination: 255.255.255.255). The packet is captured on interface 0.

No.	Time	Source	Protocol	Destination	Info
1...	0.0.0.0	0.0.0.0	DHCP	255.255.255.255	DHCP Discover - Transaction ID 0x00000000

File(F) Edit(E) View(V) Go(G) Capture(C) Analyze(A) Statistics(S) Telephony(Y) Wireless(W) Tools(T) Help(H)

dhcpr6

No.	Time	Source	Protocol	Destination	Info
9...	fe80::5689:98ff:f...	DHCPv6	ff02::1:2	Solicit XID: 0x38cdf0 CID: 00030001548998f	

<

> Frame 67: 112 bytes on wire (896 bits), 112 bytes captured (896 bits) on interface -, id 0

> Ethernet II, Src: HuaweiTe_f3:70:de (54:89:98:f3:70:de), Dst: IPv6mcast_01:00:02 (33:33:00:01:00:02)

> Internet Protocol Version 6, Src: fe80::5689:98ff:fef3:70de, Dst: ff02::1:2

> User Datagram Protocol, Src Port: 546, Dst Port: 547

> DHCPv6

The image displays two screenshots of the Wireshark network protocol analyzer interface. The top screenshot shows a packet capture of a DHCPv6 Solicit message. The bottom screenshot shows the same capture with the packet details pane expanded, displaying the DHCPv6 message structure.

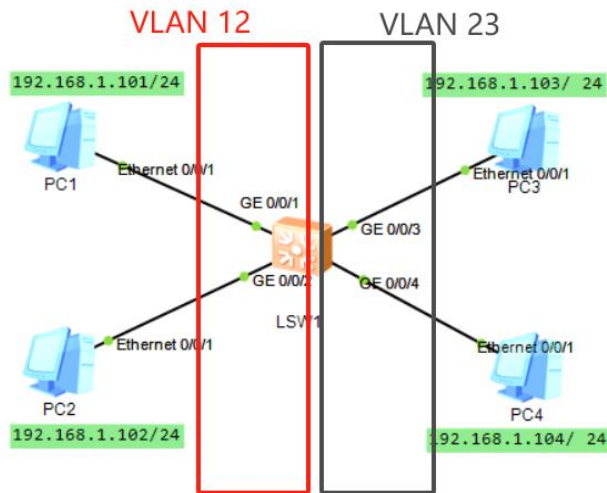
Top Screenshot:

- Menu bar: 文件(F), 编辑(E), 视图(V), 跳转(G), 捕获(C), 分析(A), 统计(S), 电话(Y), 无线(W), 工具(T), 帮助(H)
- Toolbar: Standard network analysis tools including packet list, packet details, packet bytes, and various filters.
- Packet List: Shows a single packet, 9, of type DHCPv6, source fe80::5689:98ff:f..., destination ff02::1:2.
- Packet Details: Shows the packet structure: Ethernet II, Internet Protocol Version 6, and DHCPv6.
- Packet Bytes: Shows the raw packet data in hexadecimal and ASCII.

Bottom Screenshot:

- Menu bar: 文件(F), 编辑(E), 视图(V), 跳转(G), 捕获(C), 分析(A), 统计(S), 电话(Y), 无线(W), 工具(T), 帮助(H)
- Toolbar: Same as the top screenshot.
- Packet List: Same as the top screenshot.
- Packet Details: Expanded to show the DHCPv6 message structure: DHCPv6 Solicit, XID: 0x38cdf0, CID: 00030001548998f.
- Packet Bytes: Same as the top screenshot.

Practice 13.1(1)



Build a simple network topology, with 4 PCs and a Switch. Complete the configuration on PCs and the switch.

- Configure PC as shown in the figure.
- Create vlan 12 and vlan 34.
 - make GE 0/0/1 and GE0/0/2 of Switch(LSW1) access to vlan 12;
 - make GE 0/0/3 and GE0/0/4 of Switch(LSW1) access to vlan 34

Configuration TIPS:

- step 1. create several vlans (in the system view)
 - command: # to create several vlan(s)
 - `vlan batch <vlan1-ID> <vlan2-ID> <vlann-ID>`
 - e.g. `vlan batch 12 34`
- step 2. specify the interface which access to the vlan and the vlan id(in the interface configuration view)
 - command: **port link-type access**
 - command: **port default vlan 12**

```
[Huawei-GigabitEthernet0/0/1]dis this
#
interface GigabitEthernet0/0/1
  port link-type access
  port default vlan 12
#
return
[Huawei-GigabitEthernet0/0/1]
```

```
[Huawei-GigabitEthernet0/0/3]dis this
#
interface GigabitEthernet0/0/3
  port link-type access
  port default vlan 34
#
return
[Huawei-GigabitEthernet0/0/3]
```

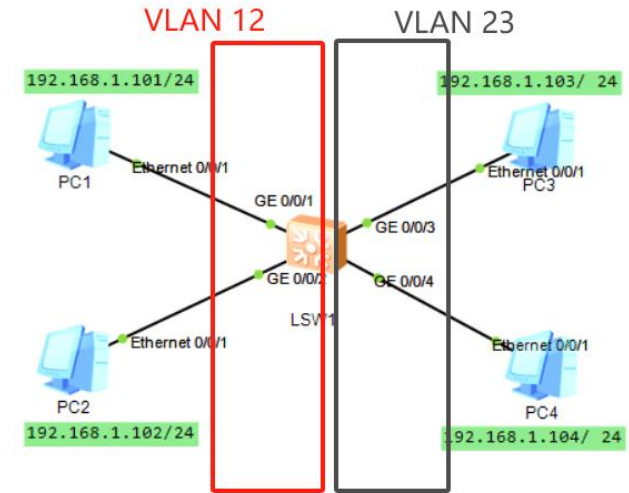
```
[Huawei-GigabitEthernet0/0/2]dis this
#
interface GigabitEthernet0/0/2
  port link-type access
  port default vlan 12
#
return
[Huawei-GigabitEthernet0/0/2]
```

```
[Huawei-GigabitEthernet0/0/4]dis this
#
interface GigabitEthernet0/0/4
  port link-type access
  port default vlan 34
#
return
[Huawei-GigabitEthernet0/0/4]
```

Practice 13.1(2)

After finish the configuration, start the following test.
capture packets for observation, and fill out the following form. Provide an explanation and clarification of the test results.

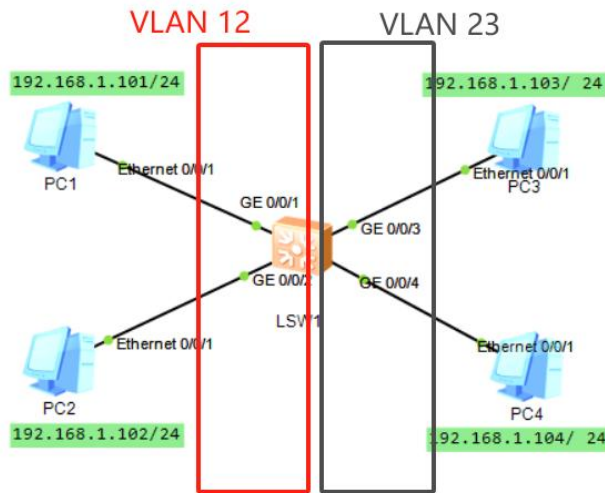
TIPS: using “dis play vlan 1” , “dis play vlan 12” and “dis play vlan 34” to find the details about these vlan.



test	pkt send ? IF pkt send , answer the following columns of questions, otherwise fill in the corresponding columns with "-"	ONLY VLAN 1 pkt:unicast/broadcast	ONLY VLAN 1 which nodes received the pkt (PC1~PC4)	Created and configured vlan12 and vlan34 pkt:unicast/broadcast	Created and configured vlan12 and vlan34. which nodes received the pkt (PC1~PC4)
PC1 “ping” PC2	PC1 send arp request ?				
	PC2 send arp reply ?				
	PC1 send ICMP echo request ?				
	PC2 send ICMP echo reply ?				
PC1 “ping” PC4	PC1 send arp request ?				
	PC4 send arp reply ?				
	PC1 send ICMP echo request ?				
	PC4 send ICMP echo reply ?				

VLAN interface(1)-IP Address

Each VLAN has its own corresponding virtual interface, which can be configured with IP address.

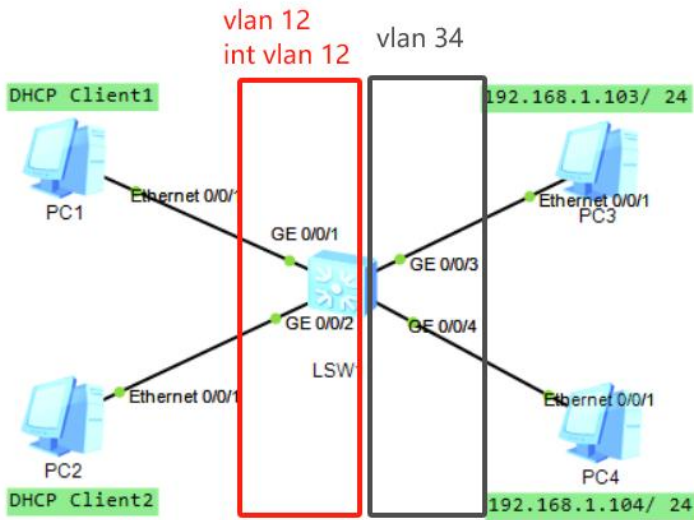


Configuration:

- step 0. create vlan (in the system view)
 - command: `#` to create one vlan
 - `vlan <vlan-ID>`
 - .e.g. **vlan 12**
- step 1. create vlan virtual interface (in the system view)
 - command: **interface vlan <vlan-ID>**
 - .e.g. **interface vlan 12**
- step 2. specify the IP address of the vlan virtual interface(in the virtual interface configuration view)
 - command:
ip address <x.x.x.x IPv4 address> <x.x.x.x subnet mask>
 - .e.g. : **ip address 192.168.1.1 255.255.255.0**
 - .e.g. : **ip address 192.168.1.1 24**

```
[Huawei]int vlan 12
[Huawei-Vlanif12]ip addr 192.168.1.1 24
```

VLAN interface(2)-DHCP server



Apply the DHCP service on the virtual interface

Configuration:

- step 1. enable dhcp service <in the system view>
 - command : **dhcp enable**
- step 2. enter the virtual interface configuration view
 - command: **interface vlan** < vlan id>
 - .e.g **interface vlan 12**
- step 3. command: **dhcp select interface**
- step 4. configure the excluded-ip-address (optional)
- step 5. configure the dns-list(optional)

NOTE: the IP address of the interface would be applied as the gateway

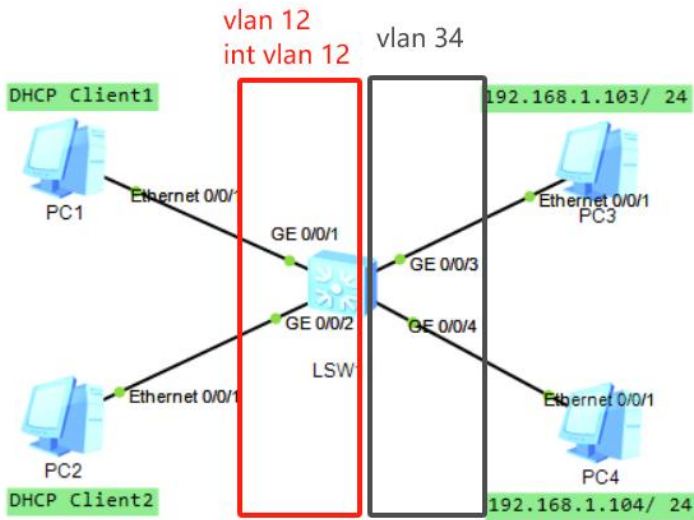
Demo3. configuration

0. The vlan 12 and vlan 34 has been create and configured in the previous steps.

1. Set PC1 and PC2 as DHCP client
2. Set virtual interface of vlan 12 as DHCP server to provide DHCP service to the VLAN 12.

```
[Huawei-Vlanif12]dis this
#
interface Vlanif12
 ip address 192.168.1.1 255.255.255.0
 dhcp select interface
 dhcp server excluded-ip-address 192.168.1.100 192.168.1.254
 dhcp server dns-list 114.114.114.114
#
return
```


Practice 13.2(1)



Before do the configuration on the IP address of the vlan interface, record the MAC address table and Routing table of the switch.

Complete the configuration on the previous page (configure the IP address of interface vlan 12 as 192.168.1.1/24, enable DHCP server on the interface), record the MAC address table and Routing table of the switch again.

Q1. What are the changes in the MAC address table and Routing table of the switch before and after configuring the IP address of the vlan interface?

Q2. Is there any new routing-entry about the vlan interface 12, is it static, direct or generated by Routing protocol?

Conduct the following tests, and answer the questions.

Test 1: Run the “ipconfig /renew” command on the PC and initiate a DHCP request message to obtain the relevant configuration. Q3. What’s the IP address, subnet mask and gateway on PC1 and PC2?

Tests 2: Initiate “ping” test from PC3 to PC1.

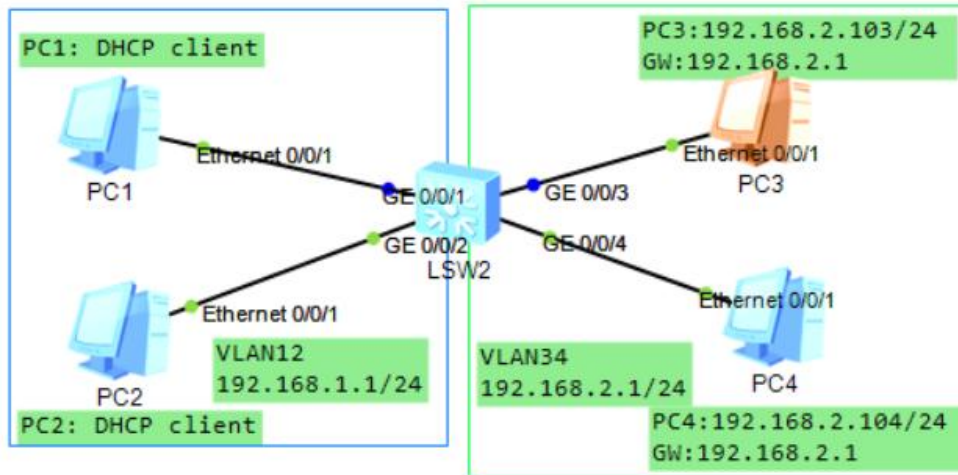
Q4. would the ARP request reach PC1?

Q5. would ICMP request reach PC1? Q6. would ICMP reply reach PC3?



NOTES: S5700 is used in this practice

Practice 13.2(2)



Make the following changes to the configuration of this question:

1. **Interface vlan34:** 192.168.2.1/24
2. **PC3:** 192.168.2.103/24, with a default gateway of 192.168.2.1
3. **PC4:** 192.168.2.104/24, with a default gateway of 192.168.2.1.

Q1. What are the changes in the Routing table of the switch before and after new configurations on the page?

Conduct the following tests, Capture pkts on GE0/0/1 and GE0/0/3 of the switch, and answer the questions.

Tests 3: Initiate “ping” test from PC3 to PC1.

Q7. Would PC3 send ARP request? what’s the value of “target IP Address” filed in the ARP request?

Q8. Would PC1 received ARP request? what’s the value of “target IP Address” filed in the ARP request?

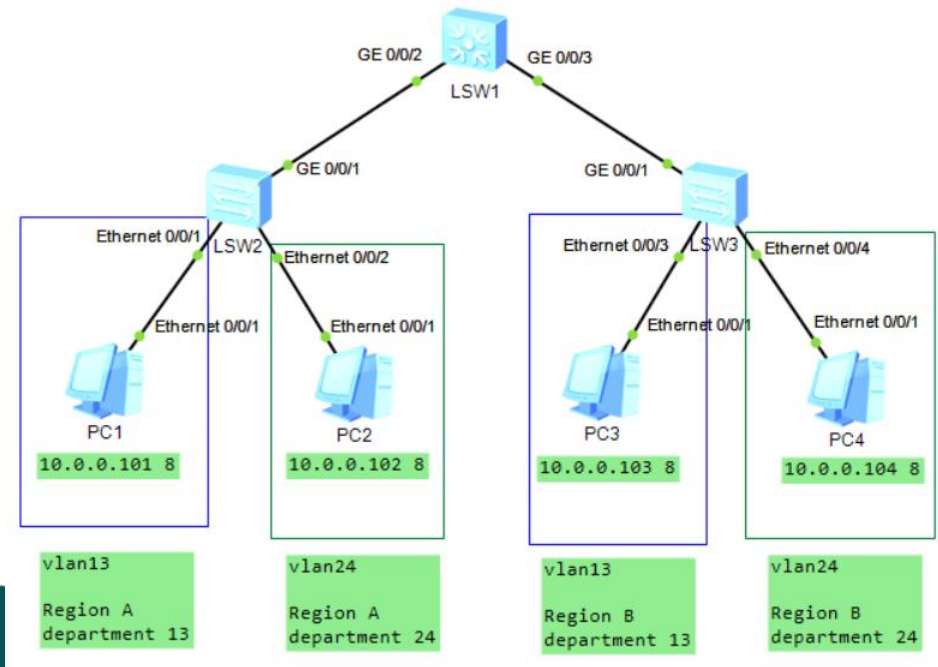
Q9. Would ICMP request reach PC1? Q10. Would ICMP reply reach PC3?

Interface on Swith: port link-type(1)

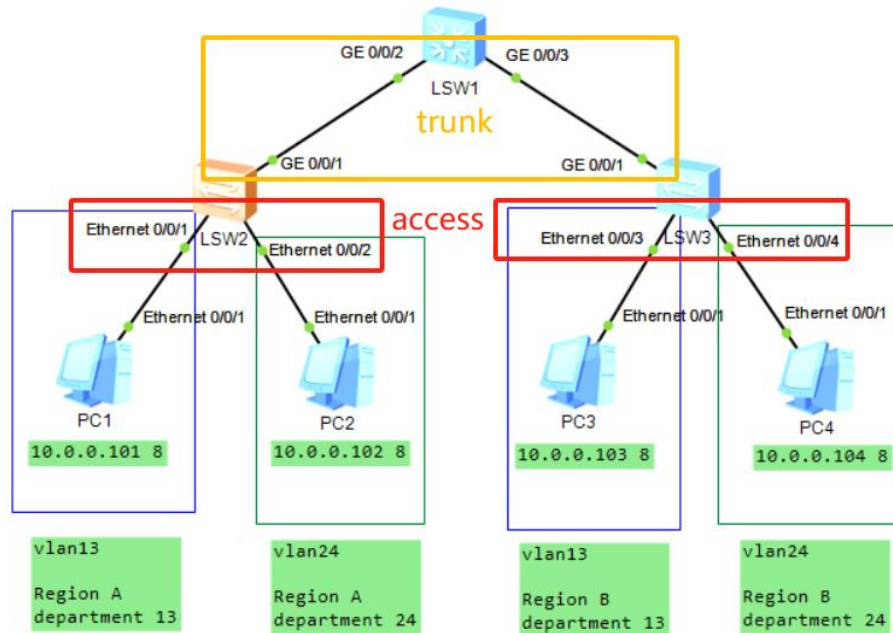
- **Access Link:**
 - A link used to connect a **user host to a switch**. Generally, the host does not need to know which VLAN it belongs to, and the host hardware usually does not recognize the frame with VLAN tag. Therefore, the frames sent and received by the host are untagged frames.
- **Trunk Link:**
 - Used for **interconnection between switches or connection between switches and routers**.
 - The trunk link can carry multiple different VLAN data. When the data frame is transmitted on the trunk link, the devices at both ends of the trunk link need to be able to identify which VLAN the data frame belongs to, so the frames transmitted on the trunk link are tagged frames.

Demo 4-1. Build a network topology as shown in the right figure:

- there are three switches in a local area network, among which LSW1 is the aggregation layer switch, LSW2 and LSW3 are the access layer switches.
 - There are two VLANs(vlan13 , vlan24) in the network, corresponding to two different departments.
 - **Terminals within department COULD access each other, but departments CANNOT access each other.**
 - The PC configuration is shown in the figure.
- Complete the configurations on the three switches .



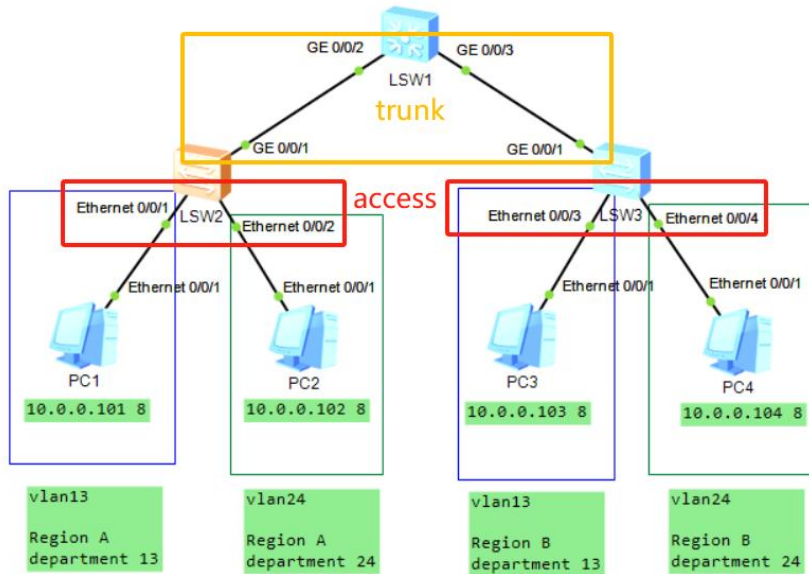
Interface on Swith: port link-type(2)



Analysis:

- 1) all the switches need to create the two vlans (vlan 13, vlan24)
- 2) the link-type of the interfaces connects to the terminal should be access, and specify the default vlan of the interface
- 3) the link-type of the interconnection between switches should be trunk
- 4) to make terminal in regin B of vlan 24 be reachable from the terminal in regin A of vlan 24, on trunk link between LSW1 and LSW2, LSW1 and LSW3, vlan 24 should be allowed to pass
- 5) to make termianl in regin B of vlan 13 be reachable from the terminal in regin A of vlan 13, on trunk link between LSW1 and LSW2, LSW1 and LSW3, vlan 13 should be allowed to pass

Interface on Swith: port link-type(3)



Configuration:

1) all the switches need to create the two vlans (vlan 13, vlan24)

- command: `vlan <vlan-id>` (in system view)
- .e.g. **vlan 13**

2) the link-type of the interfaces connects to the terminal should be “access” and specify the default vlan of the interface

- command: `port link-type access` (in interface view)
- .e.g. **port link-type access**
- command: `port default vlan <vlan-id>`
- .e.g. **port default vlan 13**

3) the link-type of the interconnection between switches should be “trunk”

- command: `port link-type trunk` (in interface view)
- .e.g. **port link-type trunk**

4) on trunk link between LSW1 and LSW2, LSW1 and LSW3, vlan 13 and vlan 24 should be allowed to pass

- command: `port trunk allow-pass <vlanx-id> <vlany-id>`
- .e.g. **port trunk allow-pass vlan 13 vlan24**

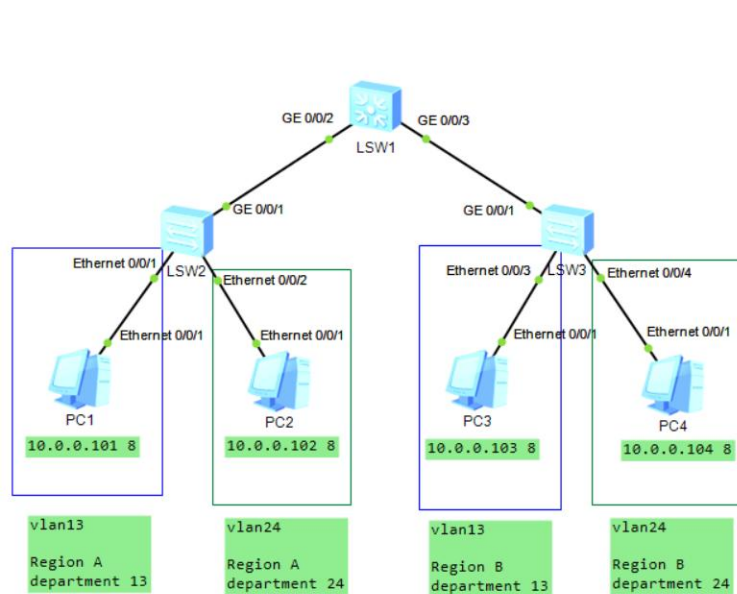
```
LSW1
[Huawei-GigabitEthernet0/0/2]dis this
#
interface GigabitEthernet0/0/2
 port link-type trunk
 port trunk allow-pass vlan 13 24
#
return
[Huawei-GigabitEthernet0/0/2]int gi0/0/3
[Huawei-GigabitEthernet0/0/3]dis this
#
interface GigabitEthernet0/0/3
 port link-type trunk
 port trunk allow-pass vlan 13 24
#
return
[Huawei-GigabitEthernet0/0/3]
```

```
LSW2
[Huawei-GigabitEthernet0/0/1]dis this
#
interface GigabitEthernet0/0/1
 port link-type trunk
 port trunk allow-pass vlan 13 24
#
return
[Huawei-GigabitEthernet0/0/1]int eth0/0/1
[Huawei-Ethernet0/0/1]dis this
#
interface Ethernet0/0/1
 port link-type access
 port default vlan 13
#
return
[Huawei-Ethernet0/0/1]int eth 0/0/2
[Huawei-Ethernet0/0/2]dis this
#
interface Ethernet0/0/2
 port link-type access
 port default vlan 24
#
return
[Huawei-Ethernet0/0/2]
```

```
LSW3
[Huawei]int gi0/0/1
[Huawei-GigabitEthernet0/0/1]dis this
#
interface GigabitEthernet0/0/1
 port link-type trunk
 port trunk allow-pass vlan 13 24
#
return
[Huawei-GigabitEthernet0/0/1]int eth0/0/3
[Huawei-Ethernet0/0/3]dis this
#
interface Ethernet0/0/3
 port link-type access
 port default vlan 13
#
return
[Huawei-Ethernet0/0/3]int eth0/0/4
[Huawei-Ethernet0/0/4]dis this
#
interface Ethernet0/0/4
 port link-type access
 port default vlan 24
#
return
[Huawei-Ethernet0/0/4]
```


Interface on Switch: port link-type(4)

Demo 4-2. tests to check if terminals within department COULD access each other, while departments CANNOT access each other.



```
PC1
Basic Config Command MCPacket UdpPacket Console
PC>ping 10.0.0.103 -c 2
Ping 10.0.0.103: 32 data bytes, Press Ctrl_C to break
From 10.0.0.103: bytes=32 seq=1 ttl=128 time=94 ms
From 10.0.0.103: bytes=32 seq=2 ttl=128 time=93 ms

--- 10.0.0.103 ping statistics ---
 2 packet(s) transmitted
 2 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 93/93/94 ms

PC>ping 10.0.0.102 -c 2
Ping 10.0.0.102: 32 data bytes, Press Ctrl_C to break
From 10.0.0.101: Destination host unreachable
From 10.0.0.101: Destination host unreachable

--- 10.0.0.102 ping statistics ---
 2 packet(s) transmitted
 0 packet(s) received
100.00% packet loss

PC>ping 10.0.0.104 -c 2
Ping 10.0.0.104: 32 data bytes, Press Ctrl_C to break
From 10.0.0.101: Destination host unreachable
From 10.0.0.101: Destination host unreachable

--- 10.0.0.104 ping statistics ---
 2 packet(s) transmitted
 0 packet(s) received
100.00% packet loss
```

```
PC2
Basic Config Command MCPacket UdpPacket Console
PC>ping 10.0.0.104 -c 2
Ping 10.0.0.104: 32 data bytes, Press Ctrl_C to break
From 10.0.0.104: bytes=32 seq=1 ttl=128 time=94 ms
From 10.0.0.104: bytes=32 seq=2 ttl=128 time=78 ms

--- 10.0.0.104 ping statistics ---
 2 packet(s) transmitted
 2 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 78/86/94 ms

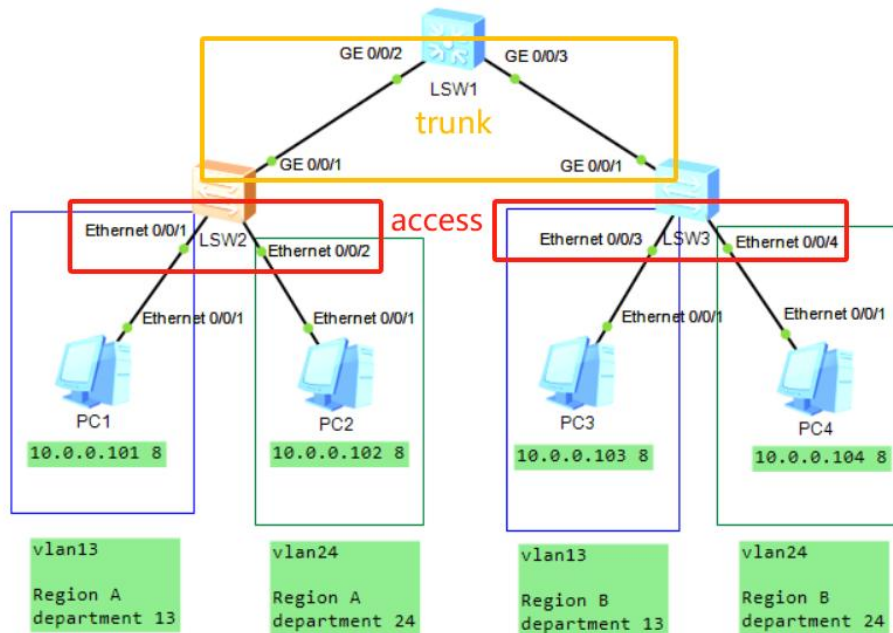
PC>ping 10.0.0.103 -c 2
Ping 10.0.0.103: 32 data bytes, Press Ctrl_C to break
From 10.0.0.102: Destination host unreachable
From 10.0.0.102: Destination host unreachable

--- 10.0.0.103 ping statistics ---
 2 packet(s) transmitted
 0 packet(s) received
100.00% packet loss

PC>ping 10.0.0.101 -c 2
Ping 10.0.0.101: 32 data bytes, Press Ctrl_C to break
From 10.0.0.102: Destination host unreachable
From 10.0.0.102: Destination host unreachable

--- 10.0.0.101 ping statistics ---
 2 packet(s) transmitted
 0 packet(s) received
100.00% packet loss
```

Practice 13.3



1. Compare the network interoperability between practice 13.2-2(page 14) and Demo4(page15-18).

What technologies are used in each demo to enable interoperability between all terminal nodes in the network?

2. Based on the Demo4, modify the configuration to ONLY allow VLAN 24 messages to pass through the trunk link.

Which nodes in the test network are interoperable ?

Show the configuration on switches and the testing result.